

Assembler Control_Almaz

Program blocks

Main [OB1]

Main Properties

General

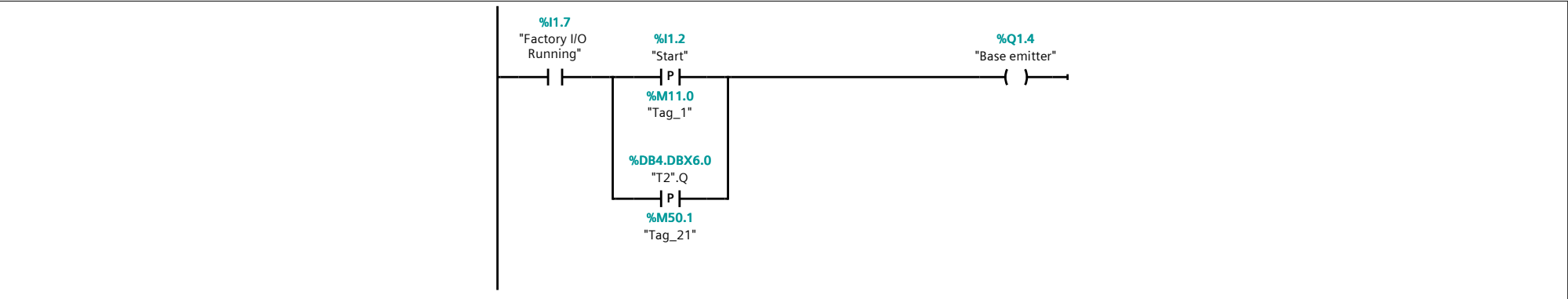
Name	Main	Number	1	Type	OB	Language	LAD
Numbering	Manual						

Information

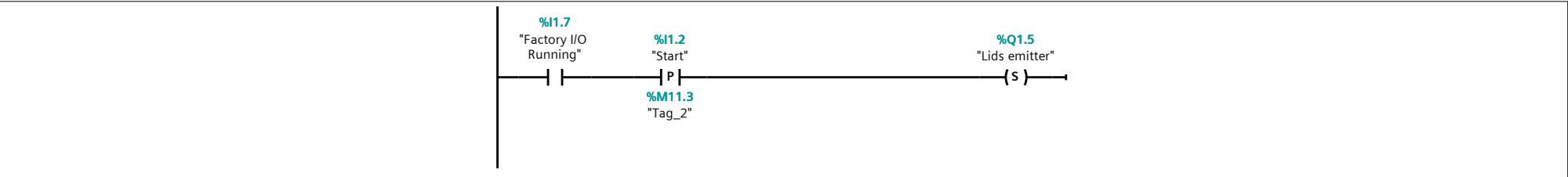
Title	"Main Program Sweep (Cycle)"	Author		Comment		Family	
Version	0.1	User-defined ID					

Name	Data type	Offset	Default value	Comment
▼ Temp				
OB1_EV_CLASS	Byte	0.0		Bits 0-3 = 1 (Coming event), Bits 4-7 = 1 (Event class 1)
OB1_SCAN_1	Byte	1.0		1 (Cold restart scan 1 of OB 1), 3 (Scan 2-n of OB 1)
OB1_PRIORITY	Byte	2.0		Priority of OB Execution
OB1_OB_NUMBR	Byte	3.0		1 (Organization block 1, OB1)
OB1_RESERVED_1	Byte	4.0		Reserved for system
OB1_RESERVED_2	Byte	5.0		Reserved for system
OB1_PREV_CYCLE	Int	6.0		Cycle time of previous OB1 scan (milliseconds)
OB1_MIN_CYCLE	Int	8.0		Minimum cycle time of OB1 (milliseconds)
OB1_MAX_CYCLE	Int	10.0		Maximum cycle time of OB1 (milliseconds)
OB1_DATE_TIME	Date_And_Time	12.0		Date and time OB1 started
Constant				

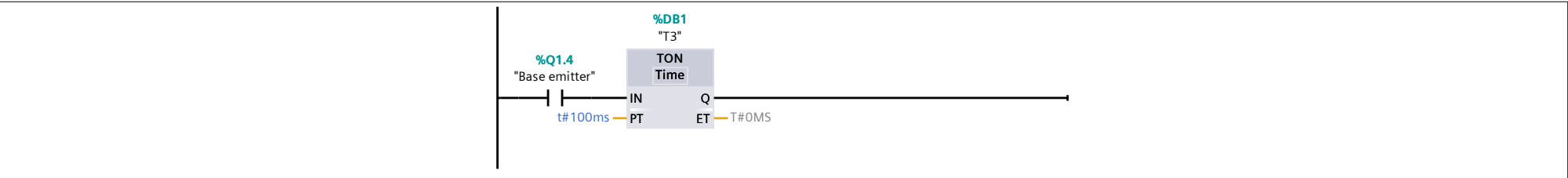
Network 1:



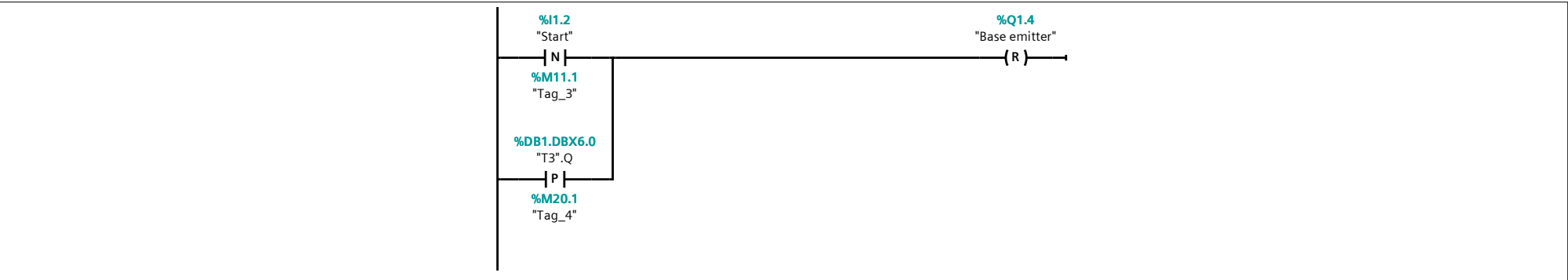
Network 2:



Network 3:



Network 4:



Network 5:

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	%I1.2 "Start" N %M11.1 "Tag_3" %Q1.5 "Lids emitter" (R)	
Network 6:		
	%I1.2 "Start" P %M0.0 "Tag_5" %DB4.DBX6.0 "T2".Q N %M11.7 "Tag_22" %Q0.3 "Lidsconveyor" (S)	
Network 7:		
	%I1.2 "Start" P %M2.6 "Tag_6" %Q1.0 "Pos. raise (bases)" P %M2.4 "Tag_23" %DB4.DBX6.0 "T2".Q N %M12.0 "Tag_24" %Q0.6 "Bases conveyor" (S)	
Network 8:		
	%I0.3 "Lid at place" N %M0.1 "Tag_7" %Q0.3 "Lidsconveyor" (R) %Q0.4 "Clamp lid" (S)	
Network 9:		
	%I0.6 "Base at place" N %M0.2 "Tag_8" %DB4.DBX6.0 "T2".Q P %M12.6 "Tag_25" %Q0.6 "Bases conveyor" (R)	
Network 10:		
	%I0.6 "Base at place" N %M12.3 "Tag_9" %Q0.7 "Clamp base" (S)	
Network 11:		

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%I0.4
"Lid clamped"

P

%Q0.4
"Clamp lid"

(R)

%M0.3
"Tag_10"

Network 12:

%I0.7
"Base clamped"

P

%Q0.7
"Clamp base"

(R)

%M0.4
"Tag_11"

Network 13:

%I0.4
"Lid clamped"

P

%M2.0
"Tag_12"

%Q0.1
"Move Z"

%Q0.2
"Grab"

%Q0.0
"Move X"

%I0.0
"Moving X"

N

%M1.1
"Tag_26"

%Q0.1
"Move Z"

(S)

Network 14:

%Q0.1
"Move Z"

%I0.1
"Moving Z"

P

%Q0.2
"Grab"

(S)

%M0.6
"Tag_13"

Network 15:

%Q0.1
"Move Z"

%Q0.2
"Grab"

TON
Time

IN

PTt#3s

Q

ETt#0MS

%DB2
"T0"

Network 16:

%DB2.DBX6.0
"T0".Q

P

%M1.5
"Tag_14"

%DB3.DBX6.0
"T1".Q

P

%M2.1
"Tag_27"

%Q0.1
"Move Z"

(R)

Network 17:

%Q0.1
"Move Z"

%Q0.2
"Grab"

%DB2.DBX6.0
"T0".Q

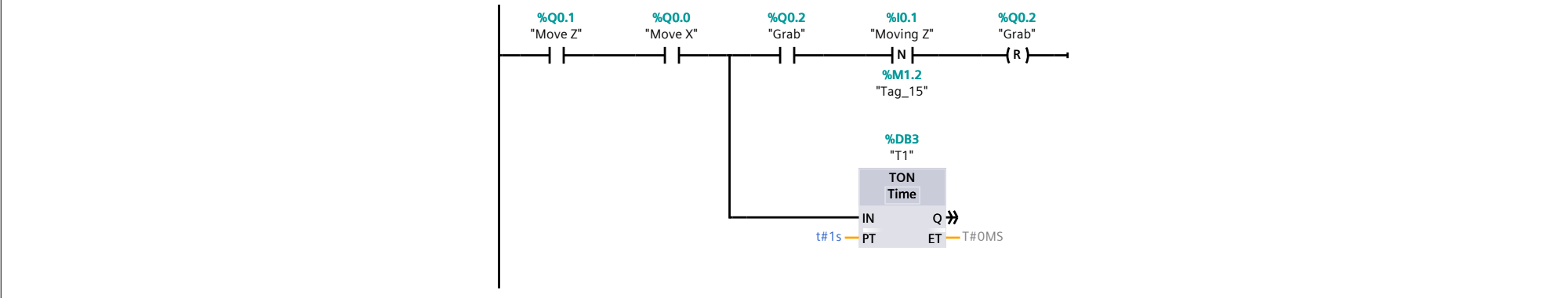
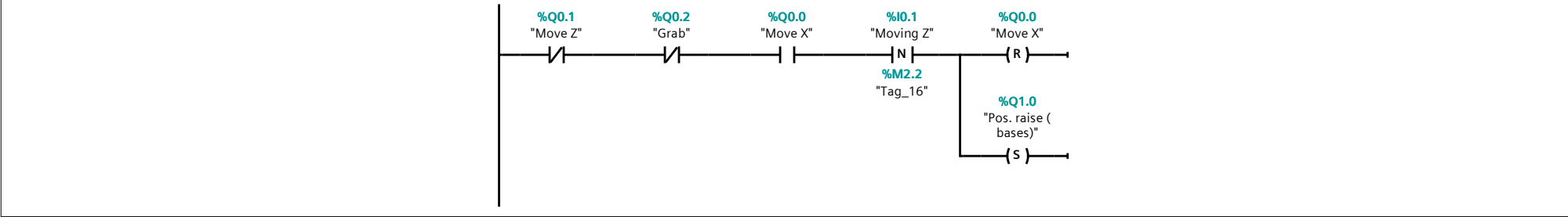
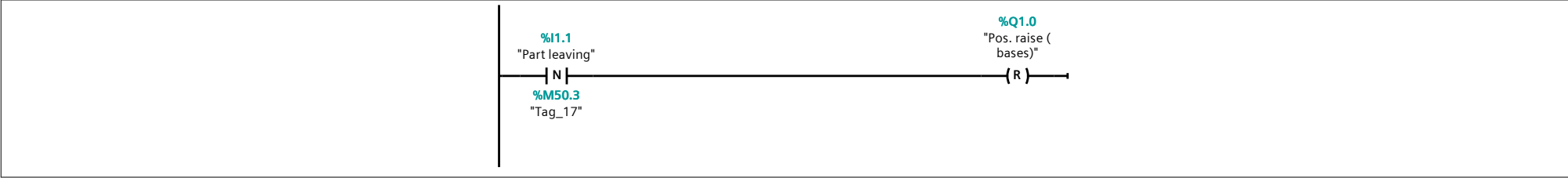
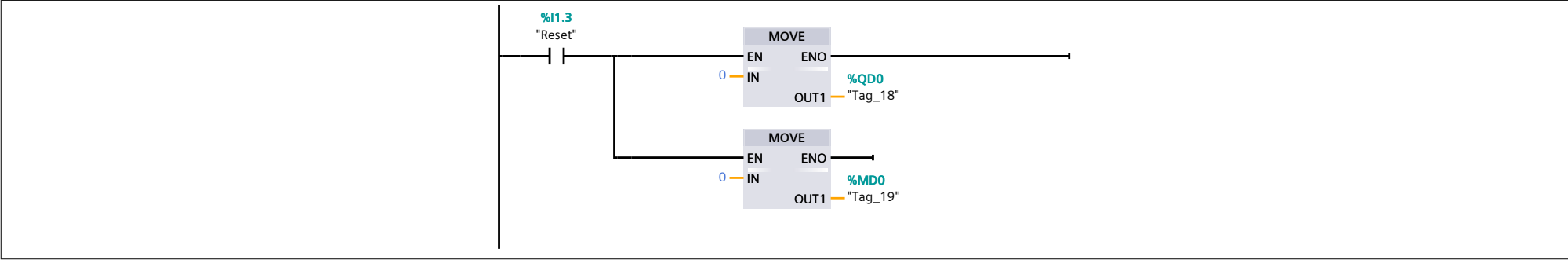
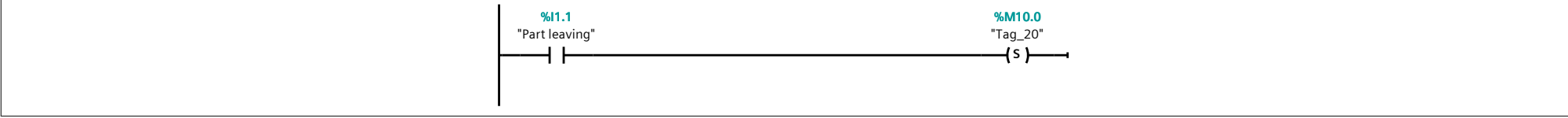
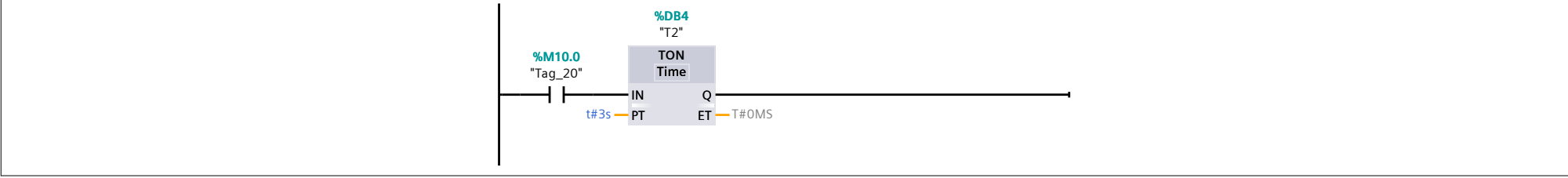
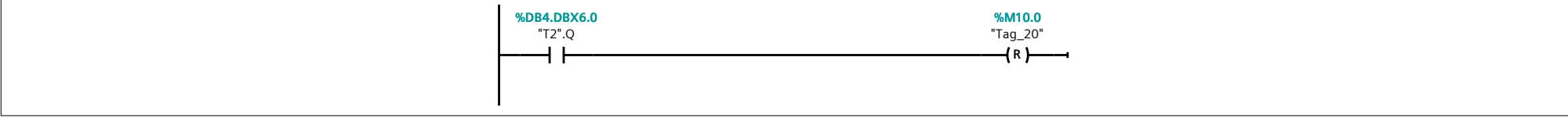
%Q0.0
"Move X"

(S)

%Q0.1
"Move Z"

(N)

Network 18:

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	 Ladder logic for Network 18: Network 18: Ladder logic with two parallel normally open contacts. The first branch contains two series contacts: %Q0.1 "Move Z" and %Q0.0 "Move X". The second branch contains two series contacts: %Q0.2 "Grab" and %I0.1 "Moving Z". Both branches lead to a coil for %Q0.2 "Grab". A timer T1 (%DB3) is connected to the first branch. The timer is a TON (Timer On Delay) with a preset time (PT) of t#1s and a time base (T#) of OMS. The timer's output (Q) is connected to the first branch.	
Network 19:		
	 Ladder logic for Network 19: Network 19: Ladder logic with two parallel normally open contacts. The first branch contains two series contacts: %Q0.1 "Move Z" and %Q0.2 "Grab". The second branch contains two series contacts: %Q0.0 "Move X" and %I0.1 "Moving Z". Both branches lead to a coil for %Q0.0 "Move X". A timer T2 (%DB4) is connected to the first branch. The timer is a TON (Timer On Delay) with a preset time (PT) of t#3s and a time base (T#) of OMS. The timer's output (Q) is connected to the first branch.	
Network 20:		
	 Ladder logic for Network 20: Network 20: Ladder logic with two parallel normally open contacts. The first branch contains two series contacts: %I1.1 "Part leaving" and %M50.3 "Tag_17". The second branch contains two series contacts: %Q1.0 "Pos. raise (bases)" and %Q1.0 "Pos. raise (bases)". Both branches lead to a coil for %Q1.0 "Pos. raise (bases)".	
Network 21:		
	 Ladder logic for Network 21: Network 21: Ladder logic with two parallel normally open contacts. The first branch contains two series contacts: %I1.3 "Reset" and %QD0 "Tag_18". The second branch contains two series contacts: %MD0 "Tag_19" and %QD0 "Tag_18". Both branches lead to a coil for %QD0 "Tag_18".	
Network 22:		
	 Ladder logic for Network 22: Network 22: Ladder logic with two parallel normally open contacts. The first branch contains two series contacts: %I1.1 "Part leaving" and %M10.0 "Tag_20". The second branch contains two series contacts: %Q1.0 "Pos. raise (bases)" and %Q1.0 "Pos. raise (bases)". Both branches lead to a coil for %Q1.0 "Pos. raise (bases)".	
Network 23:		
	 Ladder logic for Network 23: Network 23: Ladder logic with two parallel normally open contacts. The first branch contains two series contacts: %M10.0 "Tag_20" and %DB4 "T2". The second branch contains two series contacts: %Q1.0 "Pos. raise (bases)" and %Q1.0 "Pos. raise (bases)". Both branches lead to a coil for %Q1.0 "Pos. raise (bases)".	
Network 24:		
	 Ladder logic for Network 24: Network 24: Ladder logic with two parallel normally open contacts. The first branch contains two series contacts: %DB4.DBX6.0 "T2".Q and %M10.0 "Tag_20". The second branch contains two series contacts: %Q1.0 "Pos. raise (bases)" and %Q1.0 "Pos. raise (bases)". Both branches lead to a coil for %Q1.0 "Pos. raise (bases)".	

Program blocks / System blocks / Program resources

T3 [DB1]

T3 Properties

General

Name	T3	Number	1	Type	DB	Language	DB
Numbering	Automatic						

Information

Title		Author	SIMATIC	Comment		Family	IEC_TC
Version	1.0	User-defined ID	TON				

Name	Data type	Offset	Start value	Retain	Accessi-ble from HMI/OPC UA/Web API	Writ-able fro m HMI /OP C UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
▼ Input										
IN	Bool	0.0	false	True	True	True	True	False		
PT	Time	2.0	T#0MS	True	True	True	True	False		
▼ Output										
Q	Bool	6.0	false	True	True	True	True	False		
ET	Time	8.0	T#0MS	True	True	True	True	False		
InOut										
▼ Static										
STATE	Byte	12.0	16#0	True	True	True	True	False		
STIME	Time	14.0	T#0MS	True	True	True	True	False		
ATIME	Time	18.0	T#0MS	True	True	True	True	False		

Program blocks / System blocks / Program resources

T0 [DB2]

T0 Properties												
General												
Name	T0	Number	2	Type	DB				Language	DB		
Numbering	Automatic											
Information												
Title		Author	SIMATIC		Comment					Family	IEC_TC	
Version	1.0	User-defined ID	TON									
Name	Data type	Offset	Start value	Retain	Accessi-ble from HMI/OPC UA/Web API	Writ-able fro m HMI /OP C UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion			Comment
▼ Input												
IN	Bool	0.0	false	True	True	True	True	False				
PT	Time	2.0	T#0MS	True	True	True	True	False				
▼ Output												
Q	Bool	6.0	false	True	True	True	True	False				
ET	Time	8.0	T#0MS	True	True	True	True	False				
InOut												
▼ Static												
STATE	Byte	12.0	16#0	True	True	True	True	False				
STIME	Time	14.0	T#0MS	True	True	True	True	False				
ATIME	Time	18.0	T#0MS	True	True	True	True	False				

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Program blocks / System blocks / Program resources

T1 [DB3]

T1 Properties

General

Name	T1	Number	3	Type	DB	Language	DB
Numbering	Automatic						

Information

Title		Author	SIMATIC	Comment		Family	IEC_TC
Version	1.0	User-defined ID	TON				

Name	Data type	Offset	Start value	Retain	Accessi-ble from HMI/OPC UA/Web API	Writ-able fro m HMI /OP C UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion	Comment
▼ Input										
IN	Bool	0.0	false	True	True	True	True	False		
PT	Time	2.0	T#0MS	True	True	True	True	False		
▼ Output										
Q	Bool	6.0	false	True	True	True	True	False		
ET	Time	8.0	T#0MS	True	True	True	True	False		
InOut										
▼ Static										
STATE	Byte	12.0	16#0	True	True	True	True	False		
STIME	Time	14.0	T#0MS	True	True	True	True	False		
ATIME	Time	18.0	T#0MS	True	True	True	True	False		

Program blocks / System blocks / Program resources

T2 [DB4]

T2 Properties

General												
Name	T2	Number	4	Type	DB				Language	DB		
Numbering	Automatic											
Information												
Title		Author	SIMATIC		Comment					Family	IEC_TC	
Version	1.0	User-defined ID	TON									
Name	Data type	Offset	Start value	Retain	Accessi-ble from HMI/OPC UA/Web API	Writ-able fro m HMI /OP C UA/ Web API	Visible in HMI engi-neering	Setpoint	Supervi-sion			Comment
▼ Input												
IN	Bool	0.0	false	True	True	True	True	False				
PT	Time	2.0	T#0MS	True	True	True	True	False				
▼ Output												
Q	Bool	6.0	false	True	True	True	True	False				
ET	Time	8.0	T#0MS	True	True	True	True	False				
InOut												
▼ Static												
STATE	Byte	12.0	16#0	True	True	True	True	False				
STIME	Time	14.0	T#0MS	True	True	True	True	False				
ATIME	Time	18.0	T#0MS	True	True	True	True	False				